

CodeLeap Bootcamp Syllabus

Archer Infotech, Kothrud Pune | Classroom & live-online batches | Updated September 2026

Institute: Archer Infotech, Kothrud, Pune - IT training since 2009

Website: <https://archerinfotech.in>

Contact: +91 9850 678451 | info@archerinfotech.in

Programme page: <https://archerinfotech.in/bootcamps/codeleap>

Programme Overview

A 2-month foundation program in Python, Web Development, AI Tools & Career Readiness — delivered offline at our Kothrud campus and online across Pune.

Built for students who have just cleared 12th (HSC / CBSE / ICSE) and want a meaningful head start before stepping into Engineering, BCA, BSc-IT, BCS, or any IT-related college course in Pune.

Most coding bootcamps in Pune are built for working professionals or final-year engineering students. CodeLeap is different — one of the few programs in Pune designed exclusively for 12th-pass students aged 17–19 who want to enter Engineering, IT, or Computer Science with real skills already in place. In just 2 months, students at our Kothrud campus learn the same foundations that final-year engineering students struggle with: Python programming, full web development, AI tools, GitHub, and interview-ready career skills.

Programme Details

- Course Name: CodeLeap Bootcamp
- Provider: Archer Infotech, Pune
- Duration: 2 Months (8 Weeks)
- Mode: Hybrid — Online + Offline (classroom at Kothrud, Pune)
- Eligibility: 12th Pass (any stream) — Science, Commerce, or Arts
- Prior Knowledge Required: None — absolute beginners welcome
- Total Modules: 5 — Python, Web Dev, AI, GitHub, Career
- Class Schedule: Flexible — weekday & weekend batches available
- Class Size: Small, mentor-led batches (limited seats per batch)
- Languages of Instruction: English, Hindi, Marathi
- Certification: Archer Infotech Certificate of Completion
- Location: Kothrud, Pune — Maharashtra

Who This Is For

- 12th Pass-Out Students (Science, Commerce, or Arts) preparing for Engineering, BCA, BSc-IT, BCS, MCA, or B.Tech.
- First-year college students in Pune who feel underprepared for programming subjects.
- Diploma students entering the second or third year and looking to strengthen practical IT skills.
- Students from non-IT backgrounds in Pune wanting to switch to a tech-driven career path early.
- Parents in Pune searching for the best computer course after 12th for their children before college begins.

When This Programme Is The Wrong Choice

- Working professionals or final-year students looking to switch jobs — TechReady is the programme built for that, and CodeLeap will feel slow.
- Anyone expecting a job or placement at the end. CodeLeap is a foundation programme taken before college; it has no placement component and we do not pretend otherwise.

- Students who want one specific technology in depth — a single course such as Python or JavaScript will serve you better than a five-module survey.
- Anyone who cannot commit the eight weeks. The modules build on each other, and joining from Module 3 does not work.
- Students already comfortable writing programs. If you have built and deployed something yourself, start at a course rather than here.

Why Join

Local Pune institute with a real Kothrud campus

Not a faceless online platform. Our campus in Kothrud is open for visits, demo sessions, and parent counselling — so you can meet the mentors and see student work before you enrol.

Mentor-led teaching in small batches

We deliberately keep batch sizes small. Every student gets attention, every question gets answered, and every mentor knows your name and your level by week one.

Practical-first, project-led philosophy

Every concept is taught through hands-on projects. By the end of week 2 you have written real Python; by the end of week 4 you have a deployed website; by the end of week 8 you have a GitHub portfolio recruiters will recognise.

AI tools woven into every module — not bolted on

We are one of the few coding classes in Pune that treats AI fluency as a baseline skill. ChatGPT, Claude, Gemini, NotebookLM, and GitHub Copilot are taught as everyday tools, not as a separate paid course.

Hybrid flexibility across Pune

Attend in person at our Kothrud campus or join online from Karve Nagar, Warje, Kondhwa, Hadapsar, Wakad, Aundh, Pimpri-Chinchwad, Baner, Viman Nagar, or Hinjawadi — same mentors, same curriculum.

Multilingual instruction — English, Hindi & Marathi

Concepts are explained in English and switched freely to Hindi or Marathi when something is hard. No Pune student is ever left behind because of a language barrier.

Beyond the bootcamp — alumni guidance for years

Once you finish CodeLeap you stay in the loop. Alumni get continued help with college projects, internship applications, and follow-on courses across Archer Infotech's catalogue.

Curriculum

8-Week Foundation Path

Five sequential modules — Python, Web Dev, AI, GitHub, Career Skills — taught back-to-back over 2 months.

Module 1 — Python Programming (Weeks 1–2)

Programming foundations through the world's most beginner-friendly language.

Python is the world's most beginner-friendly programming language and the foundation of modern AI, web development, and data science. This module is the cornerstone of the CodeLeap curriculum and one of the most in-demand skills for Python jobs in Pune.

- Introduction to Programming - How computers think, what programming actually is, why Python.
- Python Setup & Environment - Installing Python, using VS Code, running your first program.
- Variables & Data Types - Integers, floats, strings, booleans, type conversion.
- Operators & Expressions - Arithmetic, comparison, logical, assignment operators.
- Input & Output - Reading user input, formatting output, f-strings.
- Control Flow - if, elif, else statements, nested conditions.
- Loops - for loops, while loops, break, continue, pass, range().

- Functions - Defining functions, parameters, return values, default arguments, scope.
- Data Structures - Lists, tuples, sets, dictionaries — when and how to use each.
- String Manipulation - Slicing, methods, formatting, working with text.
- File Handling - Reading and writing .txt and .csv files.
- Error Handling - try, except, finally, common Python errors.
- Modules & Packages - Importing modules, using built-in libraries (math, random, datetime).
- Mini Capstone Project - Build a working command-line application — calculator, quiz app, expense tracker, or to-do manager.

Outcome: Students leave Module 1 able to read, write, and debug real Python programs from scratch — a skill most engineering students don't develop until 2nd or 3rd year.

Module 2 — Web Development with HTML, CSS & JavaScript (Weeks 3–4)

Build, style, and deploy your first responsive websites.

Build, style, and deploy real, working websites. This module gives students the same foundation that a junior frontend developer in Pune would need on day one of a tech job.

- How the Web Works - Browsers, servers, URLs, HTTP basics, the DOM.
- HTML5 Fundamentals - Tags, structure, semantic elements, headings, paragraphs, links.
- HTML Forms & Inputs - Text fields, dropdowns, checkboxes, validation.
- HTML Media - Images, audio, video, embedded content.
- CSS3 Essentials - Selectors, properties, the box model, colors, typography.
- CSS Layouts - Flexbox, CSS Grid, positioning, modern layout techniques.
- Responsive Design - Media queries, mobile-first design, layouts that work on every device.
- CSS Animations & Transitions - Adding life and polish to web pages.
- JavaScript Basics - Variables, data types, operators, conditionals, loops.
- JavaScript Functions & Events - Click handlers, form events, basic interactivity.
- DOM Manipulation - Changing page content with JavaScript.
- Project: Personal Portfolio Website - Students build a fully responsive personal website from scratch.
- Web Hosting & Deployment - Publishing your website on free platforms — Netlify, Vercel, GitHub Pages.

Outcome: Every student walks away with a deployed, live website they can share with friends, family, college counsellors, and future internship recruiters.

Module 3 — Artificial Intelligence & Smart Tools (Weeks 5–6)

AI literacy as a baseline skill — tools, prompts, and a working AI mini-app.

This is the most future-relevant module of the entire bootcamp. AI is no longer optional — it's a baseline skill for every modern student. CodeLeap is one of the few AI courses in Pune that teaches AI literacy at the foundation level, not just as a buzzword.

- What is Artificial Intelligence? - A clear, jargon-free introduction to AI, ML, and Generative AI.
- History & Evolution of AI - From early computing to ChatGPT, Claude, and Gemini.
- How Large Language Models (LLMs) Work - Tokens, training, predictions — explained simply.
- AI vs Machine Learning vs Deep Learning - Clearing the most common confusions.
- AI in Daily Life - How AI already powers Google, YouTube, Instagram, Netflix.
- AI Across Industries - Healthcare, finance, education, design, software development.
- Hands-On with Leading AI Tools - ChatGPT, Claude, Gemini, Perplexity, NotebookLM.
- AI for Students - Using AI to study, summarise textbooks, prepare for exams.
- Prompt Engineering — The Core Skill - Writing prompts that produce great results.
- Advanced Prompting Techniques - Role prompting, chain-of-thought, few-shot examples.
- AI for Coders - Using GitHub Copilot, ChatGPT, and Claude as a coding partner.
- AI for Content Creation - Writing, design, image generation, presentations.
- Ethical AI Use in Education - Avoiding plagiarism, citing AI, building real understanding.
- AI Hallucinations & Limitations - Knowing when not to trust AI output.
- Mini Project — Build Your Own AI-Powered Mini App - Combining AI APIs with the web skills from Module 2.

Outcome: Students will be AI-fluent before they enter their first year of college — a 2- to 3-year head start over their peers.

Module 4 — GitHub, Version Control & Portfolio Building (Week 7)

Build a public developer identity recruiters can actually find.

A developer without GitHub is invisible to recruiters. This module gives every CodeLeap student a real, public developer identity online — something most engineering students in Pune don't have until their 3rd or 4th year.

- What is Version Control? - Why every modern developer uses Git.
- Installing & Configuring Git - Setting up Git on Windows and macOS.
- Git Basics - git init, git add, git commit, git status, git log.
- Working with GitHub - Creating an account, your first repository, public vs private repos.
- Pushing Code to GitHub - git push, git pull, syncing local and remote.
- Branches & Merging - git branch, git checkout, git merge — essential team skills.
- README Files & Markdown - Writing project documentation that recruiters actually read.
- GitHub Profile Optimisation - Pinned repos, profile README, contribution graph.
- Open-Source Basics - Forking, pull requests, contributing to other projects.
- Project: Upload All Your Bootcamp Work to GitHub - Python projects, your portfolio site, the AI mini-app.
- Final Project: Deploy a Personal Portfolio Site on GitHub Pages - A professional URL students can put on every resume.

Outcome: A live, public GitHub profile with multiple real projects and a deployed personal portfolio website — the single most powerful proof-of-skill any 1st-year college student in Pune can carry into internship season.

Module 5 — Career Preparation & Soft Skills (Week 8)

Resume, LinkedIn, mock interviews, and a personalised 4-year roadmap.

Skills get you in the room. Communication, presentation, and interview confidence get you the offer. The final module turns the technical foundation built over 7 weeks into a complete, market-ready professional package.

- Resume Building for Tech Roles - What recruiters actually look for; clean, results-focused formats.
- LinkedIn Optimisation - Headline, summary, projects, skill sections, networking basics.
- Aptitude Foundations - Quantitative aptitude, logical reasoning, verbal ability.
- Coding Interview Basics - How problem-solving rounds work, what to expect.
- HR Interview Preparation - Common questions, how to talk about yourself, body language.
- Mock Interviews - Live one-on-one practice rounds with feedback.
- Communication Skills - Speaking clearly, technical English, presenting your projects.
- Email & Professional Writing - Writing emails to professors, recruiters, and mentors.
- Career Roadmap Session - Personalised 1:1 guidance on your next 4 years — engineering, internships, certifications, projects.

Outcome: Every student leaves with a polished resume, an optimised LinkedIn profile, mock-interview experience, and a clear roadmap for the years ahead.

Roles this track leads to:

- Python 3
- JavaScript
- HTML5
- CSS3
- Responsive design
- ChatGPT / Claude / Gemini
- Prompt engineering
- GitHub Copilot
- Git & GitHub
- Resume & LinkedIn
- Mock interviews
- Career roadmap

Projects You Finish With

A deployed personal website

Built through Module 2 and deployed free on Netlify, Vercel or GitHub Pages. A public URL you can put on a resume, share with a college, or show a relative who asked what you did over the holidays. Responsive, hand-written HTML and CSS with JavaScript interactivity — not a template.

- Technology: HTML5
- Technology: CSS3
- Technology: JavaScript
- Technology: Netlify / Vercel
- Technology: GitHub Pages

An AI-powered mini-application

Built in Module 3, combining an AI API with the web skills from Module 2. Small but genuinely working — the point is that you have wired a model into an interface yourself and understand what happened at each step, which very few first-year students can say.

- Technology: JavaScript
- Technology: AI APIs
- Technology: Prompt design
- Technology: Fetch and async

A public GitHub profile

Your Python exercises, portfolio site and AI mini-app, committed properly with README files that explain what each one does. This is the artefact that compounds: every college project for the next four years lands in the same profile, so by placement season you have four years of visible history rather than four weeks.

- Technology: Git
- Technology: GitHub
- Technology: Markdown READMEs
- Technology: Commit hygiene

Included With Every Track

- A live, deployed personal website - Built in Module 2, deployed for free on Netlify, Vercel, or GitHub Pages — a public URL you can put on every resume, internship form, and college application.
- A public GitHub profile with real projects - Python projects, your portfolio site, and an AI-powered mini-app — all uploaded and documented. The single most powerful proof-of-skill any 1st-year student in Pune can carry into internship season.
- An AI-powered mini-application - Built in Module 3, combining AI APIs with the web skills from Module 2 — your first taste of what modern, real-world software actually looks like.
- A polished, recruiter-ready resume - Built in Module 5 around your actual GitHub work, with the formats and language tech recruiters in Pune actually look for.
- An optimised LinkedIn profile - Headline, summary, project section, skills, and a basic networking strategy — so you start engineering with a credible online developer identity.
- Archer Infotech CodeLeap Certificate - An official Certificate of Completion issued by Archer Infotech — though the GitHub profile and live website you walk away with carry far more weight in any interview room.

Tools and Technologies

By the end of CodeLeap, students gain hands-on experience with the same tools used by working developers across Pune's IT companies in Hinjawadi, Magarpatta, Kharadi, and Baner.

Programming

- Python 3
- JavaScript

Web Stack

- HTML5
- CSS3
- JavaScript
- Responsive Design

AI Tools

- ChatGPT
- Claude
- Gemini
- Perplexity
- GitHub Copilot
- NotebookLM

Developer Tools

- VS Code
- Git
- GitHub
- GitHub Pages
- Netlify / Vercel

Productivity

- Markdown
- Basic terminal / command line

Career Outcomes

CodeLeap is a foundation program, not a placement program — but the skills you learn directly translate to real career paths in Pune's tech ecosystem and beyond.

Immediate benefits

- Strong programming foundation before your 1st-year semester even begins.
- Confidence in college coding subjects — Python, C, web programming, DBMS.
- A live GitHub portfolio that grows with you through every college project.
- Eligibility for early internships — many Pune startups hire 2nd-year students with real GitHub work.
- Hackathon-ready skills — participate in college and city-wide hackathons in Pune from year one.

Longer-term paths

- Frontend Developer (React, Vue, Next.js)
- Full-Stack Developer (MERN, Django, FastAPI)
- Python Developer
- AI / ML Engineer (with further study)
- Data Analyst / Data Scientist
- Mobile App Developer (React Native, Flutter)
- Cloud Engineer (AWS, Azure, GCP)
- DevOps Engineer
- Tech Entrepreneur / Indie Hacker

Why this matters in Pune specifically

Pune is one of India's top three IT hubs — home to Infosys, TCS, Wipro, Cognizant, Persistent Systems, BMC Software, Bajaj Finserv, Citi, and Barclays, plus a thriving startup ecosystem in Baner, Aundh, Hinjawadi, and Kharadi. Engineering colleges in Pune — COEP, VIT, MIT-WPU, PCCOE, Sinhgad, PICT, Cummins, AISSMS, PCCS — produce thousands of graduates every year. The students who stand out are the ones who started early. That is exactly what CodeLeap delivers.

Fees

CodeLeap is priced as a foundation programme rather than a career programme — it is the shortest and least expensive of our three bootcamps, which is deliberate given that most students take it in the gap before college with family support. We publish fees on request rather than on the page because batch format and mode change the number. Ask for the current schedule, or visit the Kothrud campus and ask in person.

- One-time payment at enrolment
- Instalment options — ask admissions what applies to the current batch
- Free demo class before you pay anything

Placement Support

CodeLeap does not carry a placement promise, and this section exists to say that plainly rather than to bury it. Students taking this programme are 17 to 19 and heading into a degree — the outcome that matters is arriving at college already able to build, not a job offer eight weeks from now.

What the programme does give you is the career infrastructure most students assemble in a panic in final year: a GitHub profile that will be four years old by the time you need it, a resume in the format tech recruiters actually read, and a LinkedIn presence. Archer Infotech alumni also keep access to our guidance as they progress — several CodeLeap students return for CareerCode during their degree.

The process

- Module 4: GitHub profile created, projects committed and documented
- Module 5: Resume built around your actual project work, in a tech-recruiter format
- Module 5: LinkedIn profile set up with headline, summary and project section
- After the programme: guidance on which course or track to take next during your degree
- Ongoing: alumni access to Archer Infotech mentors as your college years progress

Partner companies

- Not applicable — CodeLeap is a foundation programme without a placement component

CodeLeap or a free YouTube course — is this worth paying for?

Everything in CodeLeap exists free online, and we will say that openly. Python, HTML, CSS, JavaScript and Git are among the best-documented subjects on the internet, and a disciplined self-learner can absolutely cover this ground alone in eight weeks.

What you are paying for is the part self-study reliably fails at for a 17-year-old with a free summer: a fixed schedule, someone who notices when you stop turning up, a sequence chosen for you so you are not deciding what to learn next, and a person to ask when your code will not run — which is the exact moment most self-taught beginners quit. If you have already taught yourself something end to end and enjoyed it, save your money and keep going. If you have started three online courses and finished none, that is what this programme is for.

Prerequisites and How to Begin

There are no academic prerequisites. Science, Commerce and Arts students all take CodeLeap, and no mathematics beyond school level is used. You need a laptop, an internet connection, and eight weeks in which you are not also preparing for entrance exams — students trying to do both usually do neither well.

Parents ask us regularly whether their child needs to have shown an interest in computers already. They do not. About a third of each batch arrives having never written a line of code and having no strong opinion about whether they will enjoy it; finding out is a legitimate reason to take the programme.

Suggested first steps

- Book a free demo class — attend one session before deciding anything
- Visit the Kothrud campus if you are local; parents are welcome and most come

- Check your laptop runs Chrome and can install software (that is the whole hardware requirement)
- Create a free GitHub account so it is ready on day one
- Block the eight weeks in your calendar and check they do not collide with admission formalities

How We Compare

Most Pune coding bootcamps are built for working professionals or final-year engineering students. CodeLeap is built for 17–19-year-olds who have just cleared 12th — and the curriculum reflects that, end to end.

- At CodeLeap: Designed only for 12th pass students | Most Pune coding classes: Designed for working professionals or final-year students
- At CodeLeap: 2 months — focused & complete | Most Pune coding classes: 6–12 months — long & generic
- At CodeLeap: AI tools woven into every module | Most Pune coding classes: AI taught only as a separate paid course
- At CodeLeap: Live GitHub portfolio + deployed website | Most Pune coding classes: Certificate-only, no public portfolio
- At CodeLeap: Hybrid: online + offline at Kothrud campus | Most Pune coding classes: Online-only or classroom-only
- At CodeLeap: Mentor-led with small Pune-area batches | Most Pune coding classes: Large impersonal batches
- At CodeLeap: Affordable for students | Most Pune coding classes: Premium pricing aimed at job-switchers

Frequently Asked Questions

Is CodeLeap a good coding course for 12th pass students in Pune?

Yes — CodeLeap is specifically designed for 12th-pass students in Pune. While most coding bootcamps and Python classes in Pune target working professionals or final-year engineering students, CodeLeap is one of the few programs built around the needs, learning pace, and goals of 17–19 year-olds preparing for engineering, BCA, BSc-IT, or BCS.

I have no prior programming experience. Can I still join?

Absolutely. CodeLeap assumes zero prior coding knowledge. Module 1 starts from "what is a variable" and "how does a computer think." If you have just cleared 12th and have basic computer literacy, you are ready.

Where is the Archer Infotech campus located in Pune?

Our campus is in Kothrud, Pune, easily reachable from Karve Nagar, Warje, Bavdhan, Erandwane, Deccan, Shivaji Nagar, and most parts of central and western Pune. For students from other parts of Pune (Hadapsar, Hinjawadi, Wakad, Aundh, PCMC), we also offer online live classes as part of our hybrid model.

Is this an online or offline coding course?

Both. CodeLeap is a hybrid program — you can attend in-person at our Kothrud, Pune campus, join online live, or mix the two. All sessions are mentor-led and interactive.

Will I get a certificate after completing the bootcamp?

Yes — every student who completes the program receives an Archer Infotech CodeLeap Bootcamp Certificate of Completion. More importantly, you'll walk away with a live GitHub profile and deployed portfolio website — which carry far more weight than any certificate.

How is CodeLeap different from a regular Python class in Pune?

Most Python classes in Pune teach Python in isolation. CodeLeap teaches Python plus web development plus AI fluency plus GitHub plus career skills — the complete starter package — in the same 2 months, and at a price point built for students.

What can I do after completing CodeLeap?

Students typically use CodeLeap as a launchpad before starting B.E. / B.Tech, BCA, BSc-IT, or BCS in Pune. Many continue into specialised follow-on courses at Archer Infotech (Full-Stack Web Development, Data Science, AI/ML, Mobile App Development), and use their GitHub portfolio to apply for early internships from 1st and 2nd year of college.

Is CodeLeap available in Marathi or Hindi?

Yes. Our instructors teach in English and freely switch to Hindi or Marathi when explaining tougher concepts, so no Pune student is left behind because of a language barrier.

How much does the CodeLeap Bootcamp cost?

Course fees are highly affordable compared to mainstream coding bootcamps in Pune — designed specifically with students and families in mind. For current pricing, batch dates, and any ongoing early-bird offers, please call or WhatsApp us, or visit our Kothrud campus.

Are batch sizes small?

Yes — we deliberately keep batch sizes small to maintain mentorship quality. Seats are limited per batch and admissions are first-come, first-served.

Can parents come for a counselling session before enrolling?

Of course. We strongly encourage parents to visit our Archer Infotech Kothrud campus for a free counselling session. We'll walk you through the syllabus, show you student work, and answer every question about your child's career direction.

Is there a demo class or trial available?

Yes — we offer free demo sessions before every batch begins, so students can experience the teaching style and meet the mentors. Call or WhatsApp us to book a demo.

Next Step

Ready to take the leap?

Every year, thousands of 12th-pass students in Pune begin engineering with no real exposure to programming, no GitHub, no portfolio, and no understanding of AI. You don't have to be one of them. In just 8 focused weeks at our Kothrud campus, CodeLeap puts you 2–3 years ahead of your peers — with real Python skills, a live website, AI fluency, and a public developer identity. Limited seats per batch. Admissions for the next batch are now open.

Enquire or book a free demo session. Archer Infotech, Flat No. 12, Divyadarshan Housing Society, behind Kothrud Bus Stand Road, Kothrud, Pune 411038. Phone +91 9850 678451. Email info@archerinfotech.in. Web archerinfotech.in.